CHAPTER 7

WORKING UP A TRACK: FROM MAP TO LAPS

The value of racing theory lies in its application to real problems. An analysis of the Sebring Test Circuit reveals a planning method that works at all racetracks.

ARMED WITH THE INFORmation from the preceding chapters, you should be all set to get on the racetrack and try driving a racecar. When the day arrives, you're going to be emotionally charged, which is great—it's one of the reasons we drive racecars—but you can't let the excitement interfere with your performance.

We have often seen drivers with considerable talent rely on adrenaline rather than knowledge and skill when they move from the everyday world onto the racetrack. They lock into being aggressive and inordinately brave, and are often fast—briefly. They try too hard and think too little.

"Natural talent is so rare. In the years I've been teaching, maybe one or two drivers, out of hundreds, had wonderful natural talent. And part of that was not appreciating the consequences of their actions—and being lucky. It's possible to be fast for a while by just reacting to what happens to you, but in terms of winning a championship, they're not the guys to put your money on. The analytical, methodical, technical approach is going to get you to run up front more, complete more laps, and be in a better position to win a championship. You see it all the time—drivers who win an occasional race but crash out of a lot of them."

—Jim Pace

When it's your turn to get on a proper race course, don't get caught in this trap. The professional approach is to recognize and take risks, when you need to, but to finish races. You've got to *finish* to finish first. Fastest laps are good and pole positions are great, for sure, but ultimately, you'd like to be known as a series champion.

"WORKING UP" THE RACETRACK

Let's go through the process that a good race driver uses when planning how to outperform the rest of the field. We'll take all the knowledge you've acquired and apply it to a real racetrack: the 1.8-mile Sebring test circuit in Sebring, Florida. Sebring is used throughout the winter months as one of the most popular testing sites for professional teams from CART, IRL, Indy Light, and Trans Am , to mention a few.

Details, Details, Details

The analysis we're about to start goes well beyond what many race drivers would do. But to succeed in racing—to be the fastest of the crowd—means that you have to leave mediocrity behind. If you're content to do what the "average" racer would do and leave it at that, you will limit your success.

Our experience with the type of racing we do at Skip Barber's colors our approach. Unlike many other forms of racing, the competition in both the Formula Dodge Series and in the Barber Dodge Pro Series is among drivers competing in identically prepared, equal cars. If you're going to get an advantage over others, it comes from looking within—to driving skill. This fundamentally changes your approach from trying to get the car to go faster to making yourself, as the driver, the focal point. The competition at the front of any of our series events is so close that the difference

"One of the great challenges of driving a racecar is that nothing is ever exactly the same. You always learn. You're almost never thinking, 'Oh, this is exactly how I did it last time.' You always have to be ready to adapt and change."

—Danny Sullivan

between winning and not is a matter of a tenth of a second or two per lap. This fact drives us to turn over every rock that may yield a lap time improvement.

A Plan that Works

Our intention here is not to make you a Sebring expert but to show you a process that will increase your chances of success. Our analysis process is the same regardless of the racetrack: you can use it whenever you go to a circuit you've never driven before. It's also a good idea to go through these steps at a track you know, just to confirm that there haven't been any changes since your last visit. It's a plan that works.

Start by getting a general, broad view of the racetrack to see which way it goes and to identify its obvious highlights. Once you have an overall familiarity with the course, you can hone in on the details by applying the following steps.

1. Look closely at the line. Consider whether the corner stands alone or whether another corner will force a compromise in the classic line. By walking, pedaling or slowly driving around the course, you can begin to identify likely reference points that will help you keep the car on the best path.

2. Next, consider how you would maximize the corner exit speed by determining where and how you would begin the acceleration through the apex and away from the corner.

3. Finally, focus on each corner entry to identify the likely braking process you'll need to use.

Simply put, you're trying to decide in advance what you'll do with the steering wheel, the throttle, and the brakes in each corner. To do this, look at each turn in great detail and think about the likely choices and the potential complications that will affect your line, throttle application and type of braking.

> *"A lot of what I do today is drawing on 25 years of experience, but still when I get to a new racetrack, I'll start with the basics. I'll first of all break it down into the most important corners and think about what I'm going to have to do to get the car to work there. And then I'll work back from that and start to work on subtleties. I'll really take a close look at each corner to see if there's track features that can help, things like a camber change or a pavement change I may have missed and can use."*
>
> *—David Loring*

All of this is work you can do without driving the racecar. Getting access to the racetrack in order to drive around it slowly in a street car is easy if you're polite and diplomatic. Don't abuse the privilege by driving quickly and scaring the track management and staff. Ninety percent of what you need to learn can be accomplished at less than 30 m.p.h.

Ultimately, however, some problems can only be solved behind the wheel of the racecar. Specific reference points for braking, turn-in, apex and track-out are constantly being developed and refined in the racecar at speed, starting with the general references you found in your early tours around the track at a slower pace. Before you set out in the racecar, you should have a clear idea of the unanswered questions you'll need to work on through experimentation.

Fig. 7-1. The Sebring test circuit, part of the 12 Hour course, offers a variety of challenges. An excellent training facility, it has a wide variety of cornering speeds, braking approaches and compromises.

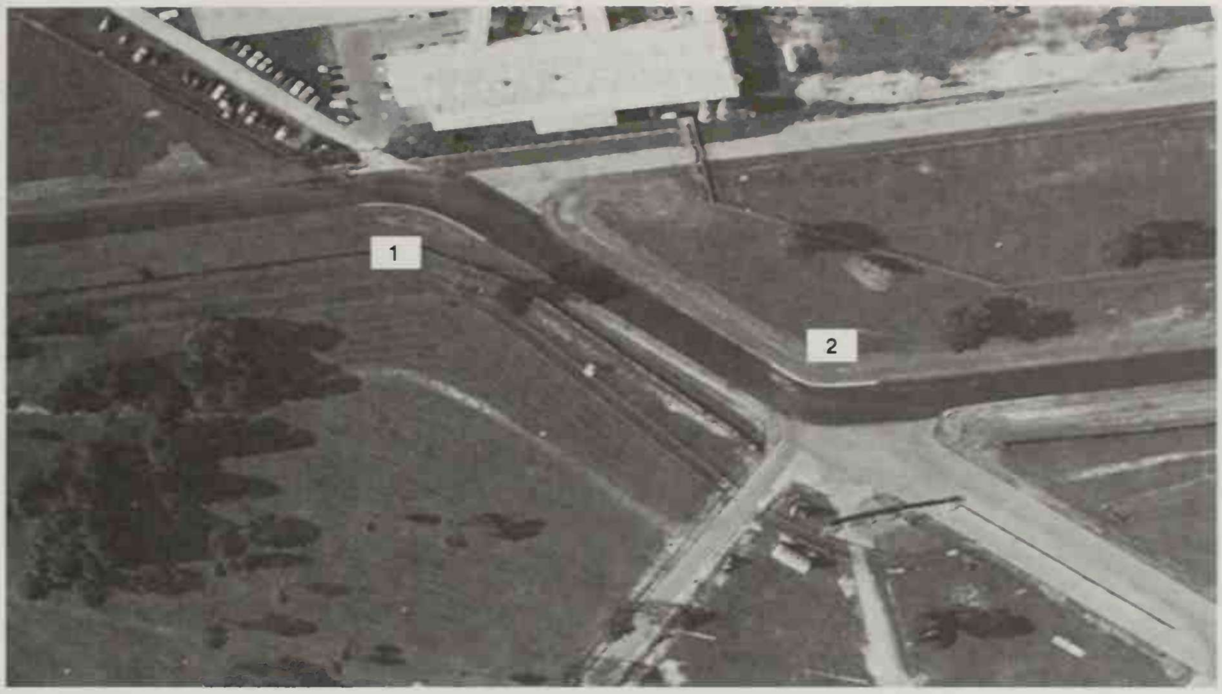

Fig. 7-2. Turns 1 and 2 make up a high-speed compromise esse: you'll determine the extent of the compromise when you're in the racecar.

Having a list of alternate approaches that you will try helps you arrive at the best conclusion more quickly.

FIRST IMPRESSIONS

The aerial photograph of the Sebring Test Circuit in Fig. 7-1 will give you an accurate idea of which way the road goes. Follow along and learn the turn numbers. They will be important in how you communicate to your crew, describing the car's, and your, behavior around the circuit. Everybody has to speak the same language to avoid miscommunication.

The first thing you will notice about Sebring is that, like most of Florida, it's absolutely flat. Changes of elevation won't be a big factor here. You'll start the lap crossing the finish line, between Turns 10 and 1.

Turns 1 and 2. It's almost a 1/4 mile run from The Hairpin up to Turn 1 so you'll expect to be going pretty fast when you get there. Turn 1 is directly affected by Turn 2 since together, they constitute a gentle esse (see Fig. 7-2). Turn 2 leads onto a straight, so of 1 and 2, it looks to be the more important corner to drive "on line."

Turn 1 may not, however, have to be completely compromised. It looks like you can use some road coming out of 1 and still get back to the right hand edge of the road for the turn-in for 2. These two corners are the fastest on the course—their arcs much more gentle than any of the others. The good news is that there is a lot of grassy run-off area so, although 1 and 2 are high-speed areas, they are relatively safe.

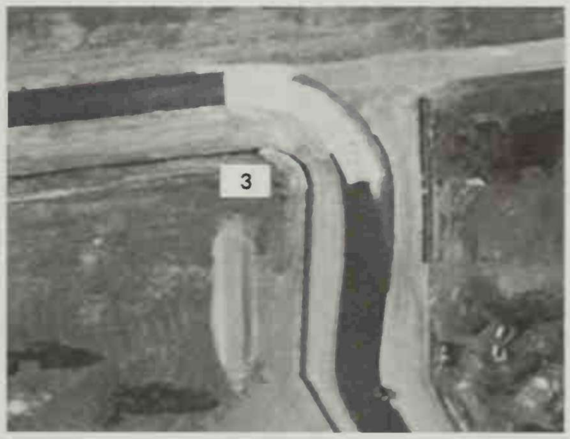

Fig. 7-3. Turn 3 is much tighter and slower than Turns 1–2 and leads to a long stretch of acceleration.

Turn 3. After tracking out of 2 all the way to the right-hand verge, you have to bend the car back to the left to get set for the approach to Turn 3 (see Fig. 7-3). You want to coax the car left, not horse it over—there should be time.

Turn 3 is a tight right-hand bend with an increasing radius: primarily concrete with lots of reference points available from the painted curbs.

Fig. 7-4. Turns 4 and 5 bend left, then right, but their radii are so large that they may be taken flat out, essentially constituting a straight.

Fig. 7-5. The gentle bend of the inside curbing of Turn 6 can be deceiving, tempting you to turn early.

Fig. 7-6. Turn 7 is a short, classic, 90-degree right.

Turns 4, 5 and 6. Turns 4 and 5 look to be just bends in the road, one left, one right, that will probably be taken flat out, accelerating as hard as possible (see Fig. 7-4). The run through 4 and 5 and up to the braking zone for Turn 6 is a long stretch of road, just as long as the straight between The Hairpin and Turn 1. Exit speed out of Turn 3 will be important.

Turn 6, another right-hander, also has entry and apex areas made of concrete but, worthy to note, the concrete ends before the track-out (see Fig. 7-5).

The approach to Turn 6 is very deceiving. The inside curbing angles gently down toward the apex while the outside edge of the racetrack is shaped like a sharp 90 degree corner. Look to the inside and the corner looks more gentle— and faster—than it does if you look toward the outside. All that road down there at the entry will tempt you to turn in too early.

Turns 7, 8 & 9. Turn 6 is followed by a very short straight leading to Turn 7, a textbook 90-degree right-hander and a very short corner (see Fig. 7-6). You're in and out quickly, but it will be quite slow—you might consider using first gear.

Fig. 7-7. Turn 9 leads onto the longest period of full throttle acceleration so you'll plan on sacrificing 8 to be sure you're on line for 9.

Fig. 7-8. The slowest corner on the racetrack is The Hairpin. If any corner in the world requires threshold braking, this is it.

Turn 8 is just a jog to the right and is going to be important mostly because it leads to Turn 9, the entry to the longest straight.

Turn 9 seems to go on and on, a long corner with smooth asphalt, well over a 90-degree change of direction, with a decreasing radius as well (see Fig. 7-7). You'll need a late apex as well as some device to keep you from getting in early and having to lift coming onto the longest straight.

The "straight" isn't. It bends constantly to the right as the road sweeps gently under the vehicle bridge, then, a quarter-mile later, still turning right, under the pedestrian bridge. This half-mile long curving straight is the section of the course where you are likely to reach the highest speed, accelerating with your foot to the floor up through the gears. The straightaway is a chance to take a breath, wiggle your fingers, check the mirrors and gauges, and let the car do the work.

Turn 10: The Hairpin. Finally, 2,000 feet after you finish Turn 9, you flash by the second bridge and the road straightens for 200 hundred yards or so where you begin braking for Turn 10 (see Fig. 7-8).

The Hairpin is tight and slow—our Formula Dodge racecars, as run in our race series, are in the mid-30 m.p.h. range at the slowest point. The corner is also followed by a 1/4-mile straight, so exit speed and throttle application to deal with wheelspin, are going to be big considerations here.

The entry to this turn is odd. Obviously, there is a lot of braking going on here, from whatever top speed the car can generate down to around 35 m.p.h. Should you have a failure, there is an escape route to the left, a route many find hard to take since their minds are focused on turning right. Fifty feet before the turn-in point, the track surface changes from asphalt to concrete. In the last 20 yards of braking zone, the road widens to the left with a two-foot wide curbing you could use to widen the arc of the corner. You can try it later to see if it upsets the handling of the car. The apex curb here is pretty steep and it probably won't help to hit it with the right front tire. The exit is smooth but sandy.

These first couple of laps, driven slowly in a street car, give you first impressions. Now it's time to get out a track map and try to get specific about the problems presented by each corner, thinking about how you're going to solve them.

FOCUS ON THE DETAILS

Take as much time as you need. Remember, problems are much easier to ponder and solve at this stage than they will be at speed. Issue Number One is the line.

Try to decide whether a corner can be considered singly or should be compromised for the next corner; and identify the more important corners—those leading onto long periods of acceleration as well as those that are long and fast. Look at each corner's shape and note changes in banking or road surface that will affect your apex. At the same time, scout out your reference points for the turn-in, apex and track-out of each corner.

Next, consider corner exits—concentrate on where and how you will likely apply the throttle.

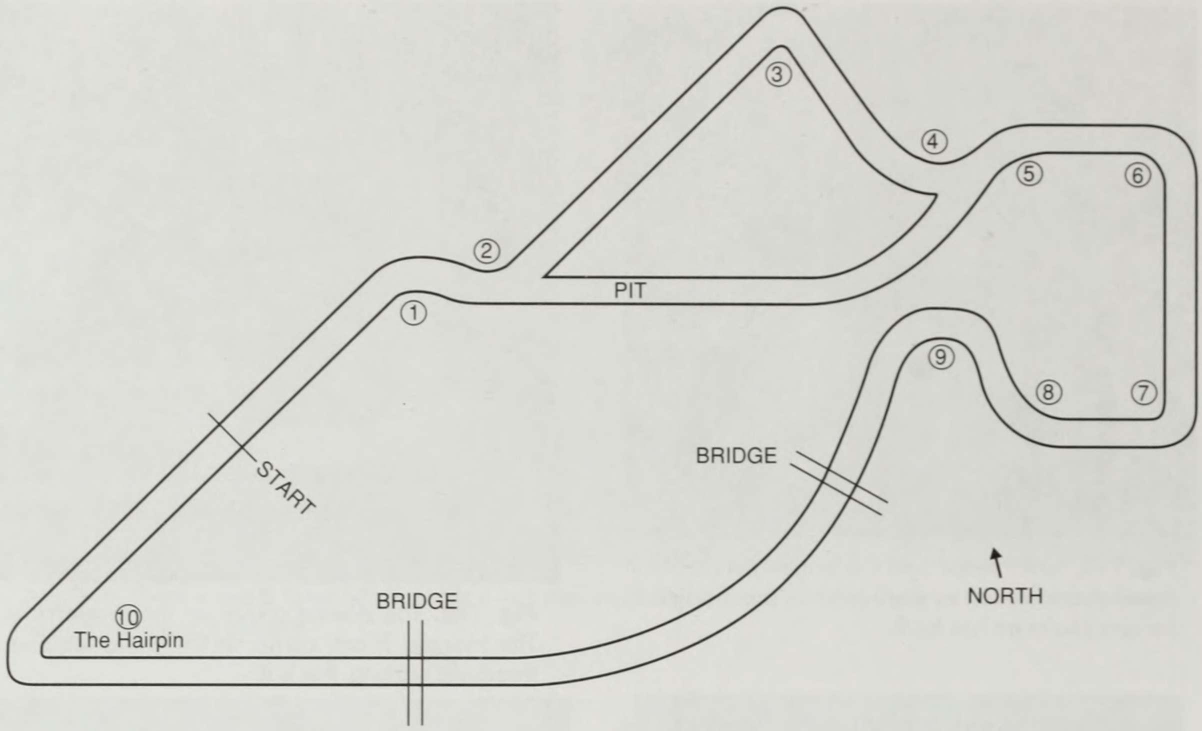

Fig. 7-9. Track maps are seldom to scale but can be useful in making driving decisions before getting in the car.

Finally, on the last 25 laps, consider the type of braking required for each corner based on speed loss and location, all the while looking for your braking reference points.

Let's apply this method to the Sebring test circuit, starting with Turns 1 and 2.

Turns 1 and 2

The Line. The obvious problem here is that you can't drive outside-inside-outside in Turn 1 and be able to get over to the turning point for the classic line through 2 (see Fig. 7-10).

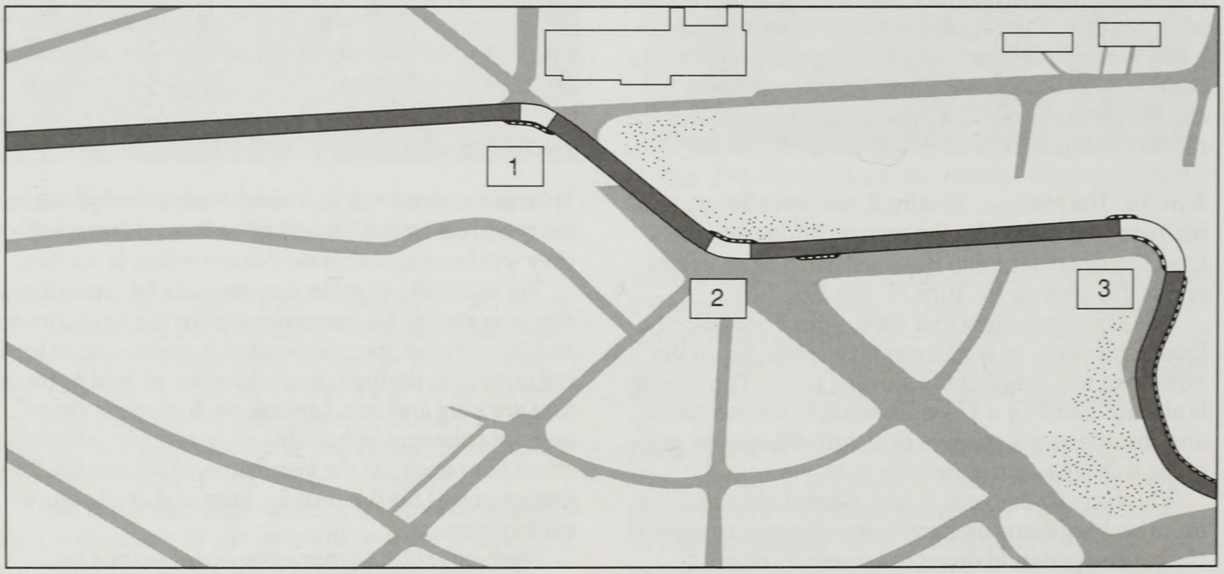

Fig. 7-10. You should plan to compromise Turn 1 in order to drive on the biggest radius through Turn 2, since the track segment from 2 to 3 is longer than Turn 1.

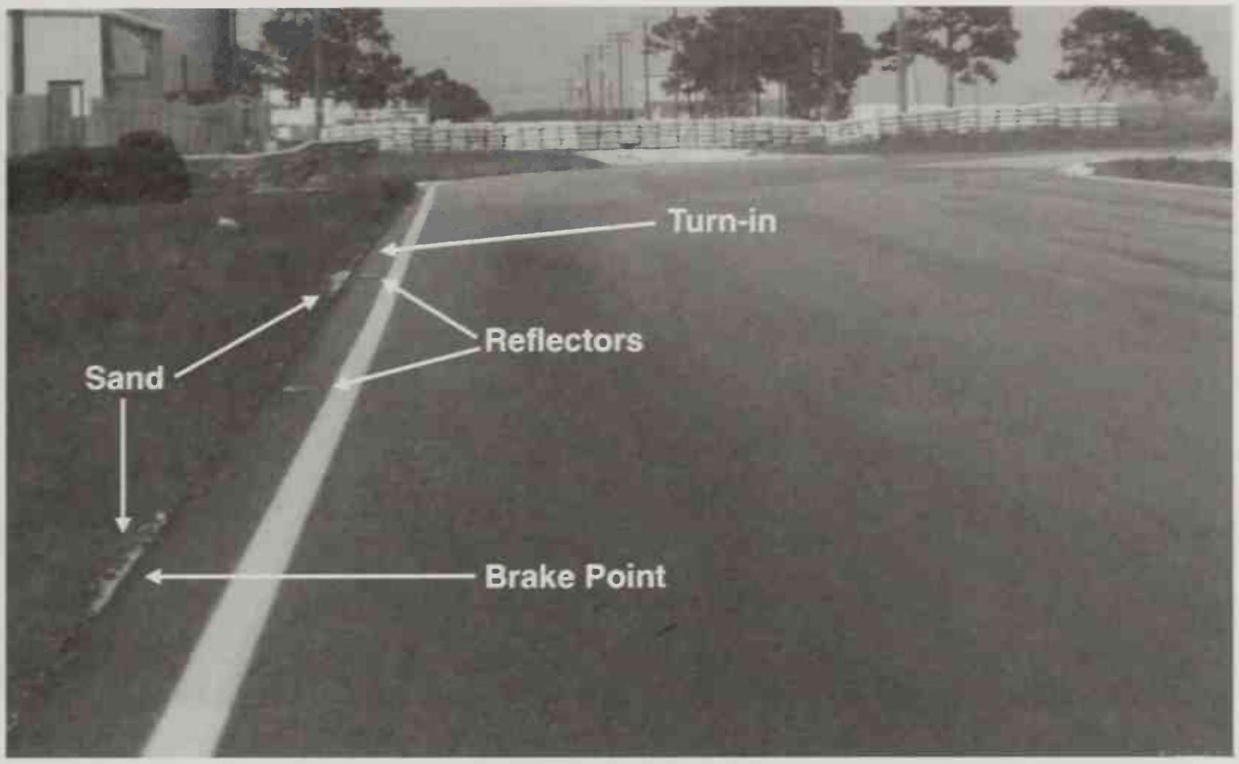

Fig. 7-11. The approach to Turn 1 has very uniform features, so you'll have to find creative reference points.

In choosing which turn is more important to drive on-line, your choice will depend upon the length of the Turn 1 track segment versus the length of the track segment from the turn-in point for 2 to the next braking zone. You'd prefer to carry the extra speed which the classic line will allow over the longest portion of racetrack. The distance from the approximate turn-in of 1, through the apex and up to the turn-in for 2 is about 450 feet. The turn-in to 2, through the apex, past the track-out and up to the braking point for 3 is almost 600 feet. You'd rather carry extra speed for 600 feet than for 450 feet, so the solution would appear to be to compromise the line through 1 to get the best shot through 2.

If Turn 2 led onto a long straight, the choice would be obvious: sacrifice the first corner for the second. When the second corner leads nowhere, you might not choose to sacrifice the first.

When it comes to compromise corners, keep in mind that the big radius advantage of the classic line is most useful when the line determines the speed. The line limits the speed through the corner when the car is operating at its cornering limit. If Turn 1 were tighter, say 30 m.p.h. slower than Turn 2, you might not be able to get the car going fast enough into 2 that it needs the big radius offered by the classic line. In this case you could compromise Turn 1 less, making its radius bigger, and be able to carry more speed through 1 *and* 2.

This isn't the case here. The car will be at its cornering limit in both 1 and 2 so the line through these turns determines the car's speed, as long as you exercise the skill to drive the car at its cornering limits in the 75–85 m.p.h. range.

Another consideration for trying to run on the biggest possible arc, even though you don't need it to accommodate limit cornering, is the plain fact that more steering wheel lock creates more tire scrub, causing drag, which resists acceleration. We'll deal with this consideration shortly.

The line difficulties in these two corners are going to involve picking up a good reference point for the turn-ins for 1 and 2. The approach to 1 has very uniform features, making it tough to pick out a brake point and turn-in point (see Fig. 7-11). The apex for Turn 1 falls in the middle of the short curb, an obvious aiming point.

At the turn-in for Turn 2 (see Fig 7-12), you'll be traveling near 80 m.p.h., just finishing your drift through 1, straightening for an instant, then committing to 2. Getting a consistent turn-in reference is going to be something you need to develop at a conservative pace and pick up with your peripheral vision since, in this especially fast corner, you want your eyes on the apex when you turn. The apex reference for Turn 2 is easier; the concrete patch in the center of the corner makes a good target (see Fig. 7-13).

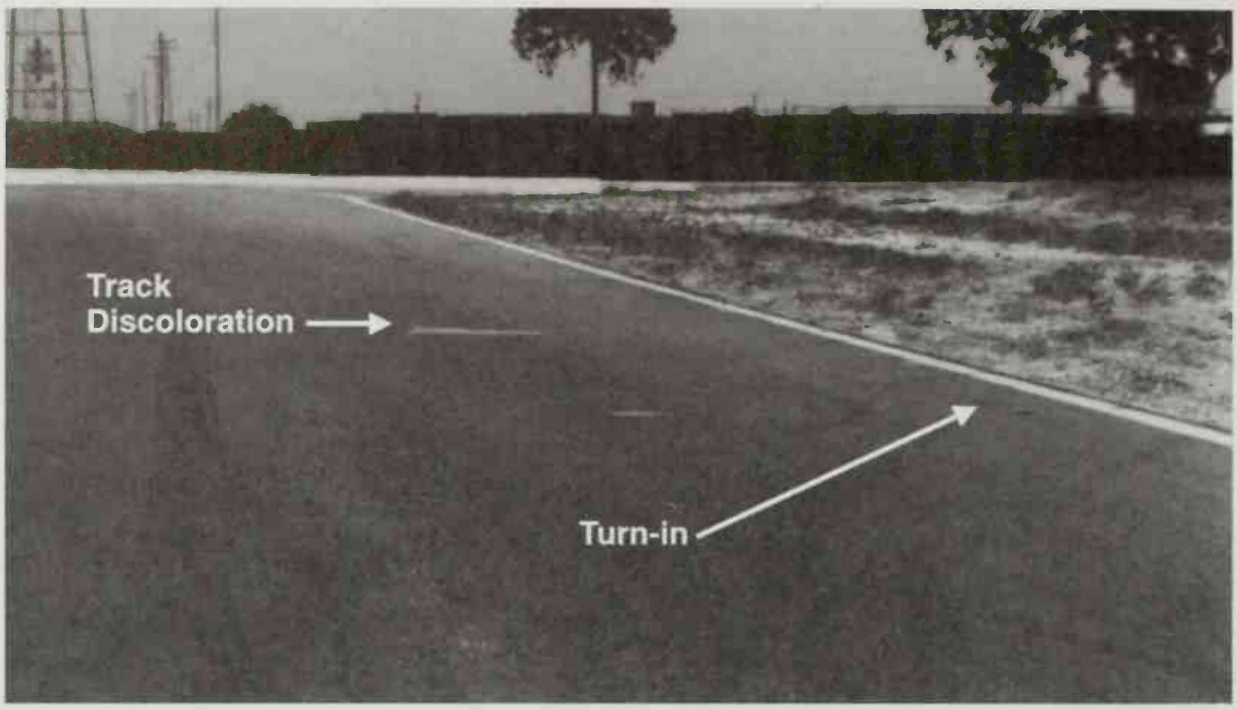

Fig. 7-12. The white discoloration at the right of the track before Turn 2 looks like it's too late for a turn-in point; vary your distance to these points as you experiment with the turn-in.

The track-out for 2 has a relatively flat, narrow curb and the adjoining dirt runoff is quite level so that, should you drop a wheel, it shouldn't create a big control problem as long as you don't try to horse the car back to the left. It's also worth inspecting how deep the drop-off on the back side of the curb has become. Sometimes, if it's been recently graded by the track maintenance crew, it will be even with the track surface. But if it's a busy race weekend, the drop-off can be like a canyon and has to be avoided at all costs.

At the apex of Turn 2, there is a big, sudden camber change which, even at low speed, feels like a big bump (see Fig. 7-13). It's bad enough that if you had the car at an excessive yaw angle and were just starting to gather it up when you hit the bump, it could trigger a spin. Unfortunately, you can't avoid it as this would

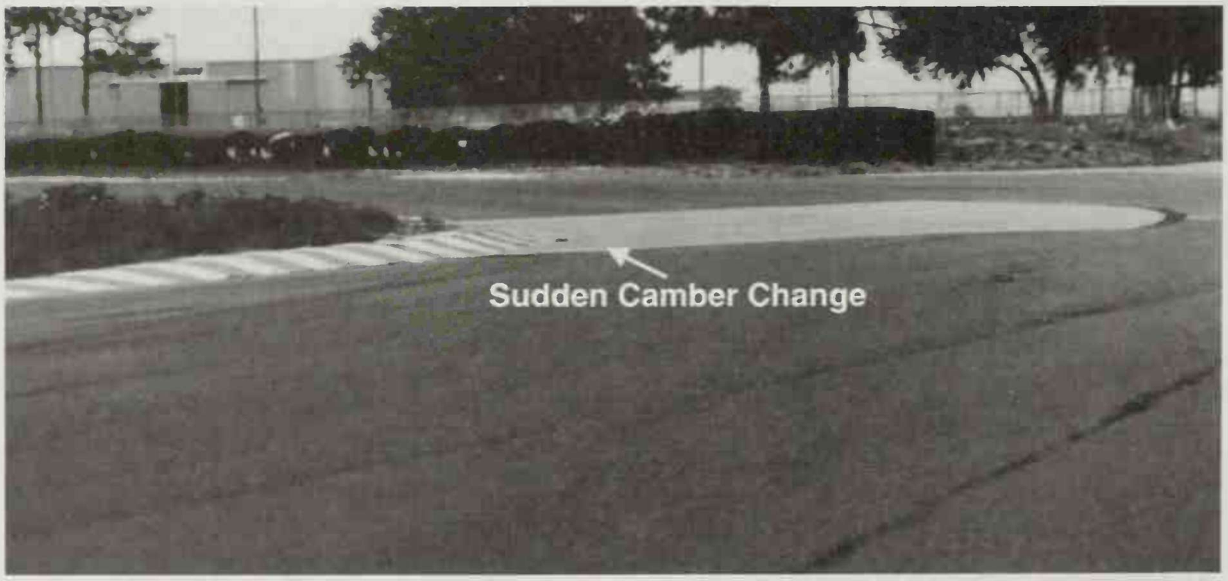

Fig. 7-13. The apex of Turn 2 falls in the center of the concrete patch. Note the sudden camber change at the apex. Track-out is at the very end of the exit curbing.

put you 10 feet wide of the apex. Just remember that if the car is out of shape approaching the apex of 2, it's important to get it settled before the apex bump.

Throttle Application. Since the two are directly connected it makes sense to take them together. Since both are fast corners—probably third gear—you won't have to worry about generating wheelspin with too much throttle. In Turn 1, you'll expect to get back to full throttle right after brake release and carry a lot of speed through the apex of 1. The trick is going to be the entry of Turn 2. You can't tell, until you're in the racecar, how much you will have to slow the car for the entry of 2. Presumably, you won't be flat on the throttle through both Turns 1 and 2. You'll have to slow the car some to get the car turned into 2. In any event, you *do* know that you need to be careful of snapping off the throttle while the car is still turning right, exiting 1.

Expect that, if you need to slow for 2, you can breathe back the throttle while finishing 1, but that any full lift or braking maneuver has to be saved for the period when the car gets straight for a bit between 1 and 2. In high-speed sweepers like these, the car will be more stable with some power on and its yaw can be controlled with small—10–20%—power changes.

Braking. Although Turn 1 comes after a long straight, it's such a gentle bend that it's doubtful that you'll have to lose a lot of speed for the corner entry. Rather than expect threshold braking, you'll most likely want to control the entry speed more accurately with a lighter braking level. This will make the load transfer more progressive and help make the car feel less upset at the corner entry. You're likely to trail the brakes into the corner a bit but, since you're keen to get back on the throttle to stabilize the speed and handling, you'll probably trail less than normal.

You won't know whether the brakes are needed in turn 2 until you try it. If they are, it will be a light application to lose just a touch of speed to enter 2. If braking isn't needed, the speed adjustment will be just a breathe of the throttle. Be careful of TTO.

The bulk of the breathe should come while the car is briefly straight between 1 and 2, and you should expect to stabilize the car with some power just as you turn toward the apex of 2.

Turns 3, 4, and 5

The Line. At first, Turn 3 (see Fig. 7-14) looks as if it can be considered all on its own. It's a tight right-hander, with little road camber to consider and a concrete surface throughout the last 2/3rds of the corner. It leads onto a sweeping acceleration section of the course which is almost a quarter mile long from the exit of 3 to the turn-in for 6, and this can have an effect on the line you choose.

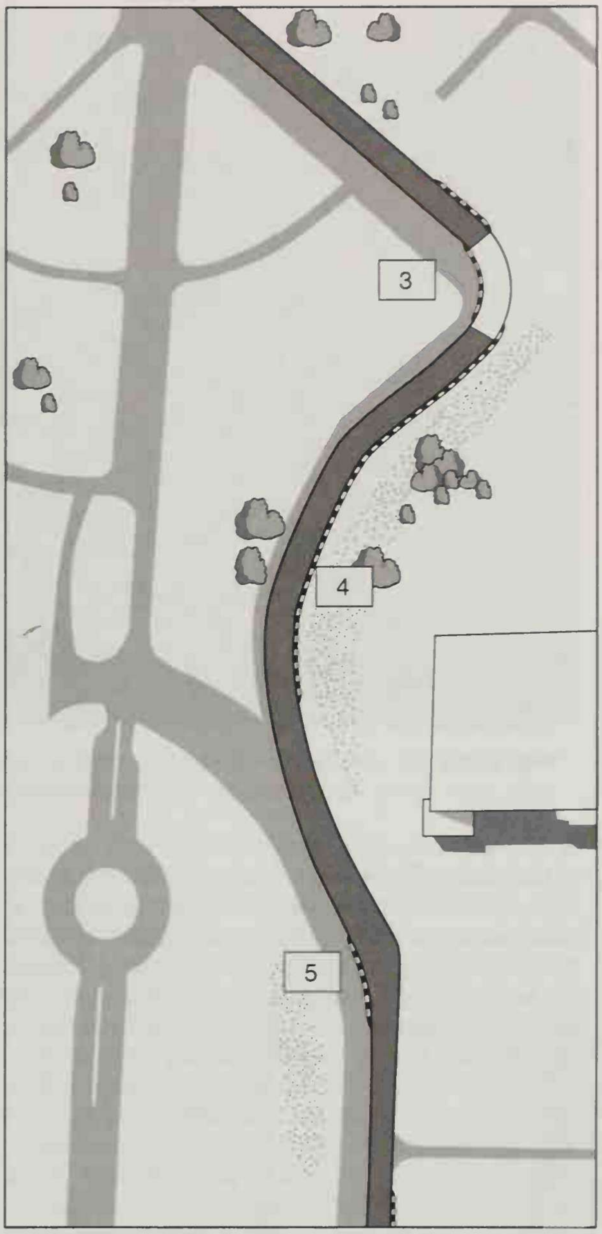

Fig. 7-14. Experimenting in the car at speed in Turn 3 will tell you whether to use all of the road at the track-out point.

The corner entry is not a problem. A curb on the left is going to be helpful in picking out a turn-in point and the striped apex and the complex of seams in the concrete will make it easy to pick an apex mark (see Fig. 7-15). The concrete section starts about halfway between the turn-in and the apex so you'll have to look for changes in grip between the two surfaces to see if it affects your line.

The biggest complication comes from deciding whether to let the car track out all the way to the edge

Fig. 7-15. The apex for Turn 3 is not a problem: the striped curb and the concrete seams will allow you to pick your apex mark easily. Also remember that the change to concrete and then back to asphalt will affect your grip.

of the road at the exit or to take a line that holds it a car width or two toward the middle of the road.

If you let the car all the way out at the exit you will then have to hug the inside edge of the road through the beginning of Turn 4. You can take this section flat out, accelerating as hard as you can, but if you start on the left you'll have to turn the steering wheel more, especially through 4. Using more steering angle will create more scrub against the road, adding resistance to acceleration. Minimizing tire scrub on the acceleration run up to Turn 6 could be worth lap time. If you decide not to track all the way out of Turn 3, the down side is that, because of the tighter radius, your speed out of 3 will be lower. You may accomplish your goal of limiting tire scrub but, since you start the acceleration run from a lower speed, it may all come out even in the end and the elapsed time from 3 to 6 will be the same, or maybe even worse. There's no way to know but to try it both ways at speed. Your plan in this case is to stick with one approach and when it's time to start looking for the last second of lap time, to experiment in Turn 3.

Tire scrub dominates the approach to Turns 4 and 5. As the car accelerates around this long left-hand bend , you want to take the line of least resistance, especially with a car with lower horsepower. Tire scrub will be a bigger factor in limiting the acceleration of a low horsepower car than it will on one with 600 or more horsepower.

In a low horsepower car you want the car to take its own smooth path, never trying to muscle it to turn more, but attempting to unwind the steering wheel—within the confines of the road, of course. Guiding the car through Turns 4 and 5 will be a matter of fingertip control of the steering wheel, letting the car have its head, nursing the car up to speed (see Fig. 7-16). You will have to be careful not to get out too far right as the outside is filled with the sand, grit and bits of used up tire rubber we call "the marbles." It can get very slick on the marbles and, even though you're not on the limit cornering through 4 and 5, the marbles can lower the traction enough to create a real problem.

The actual line through 4 will take you close to the apex curb twice as you try to create a gentle arc up to 5 (see Fig. 7-17). The first curbing is about 150 feet past the track-out of 3 and is very steep, so it's not to be hit. Come close, but try not to hit it. The car floats right, away from the curb, if the steering wheel is held at the same angle, then floats back toward the inside again getting closest about another 300 feet along. You let the car drift right again heading for the apex curbing at Turn 5. As you get to 5, the steering wheel is almost straight. You would have to actually turn the wheel right to get to the apex of 5 so you don't bother. Your path is headed diagonally toward the entry of Turn 6 and there's no reason to drive over to the apex curb for 5 simply because it's there.

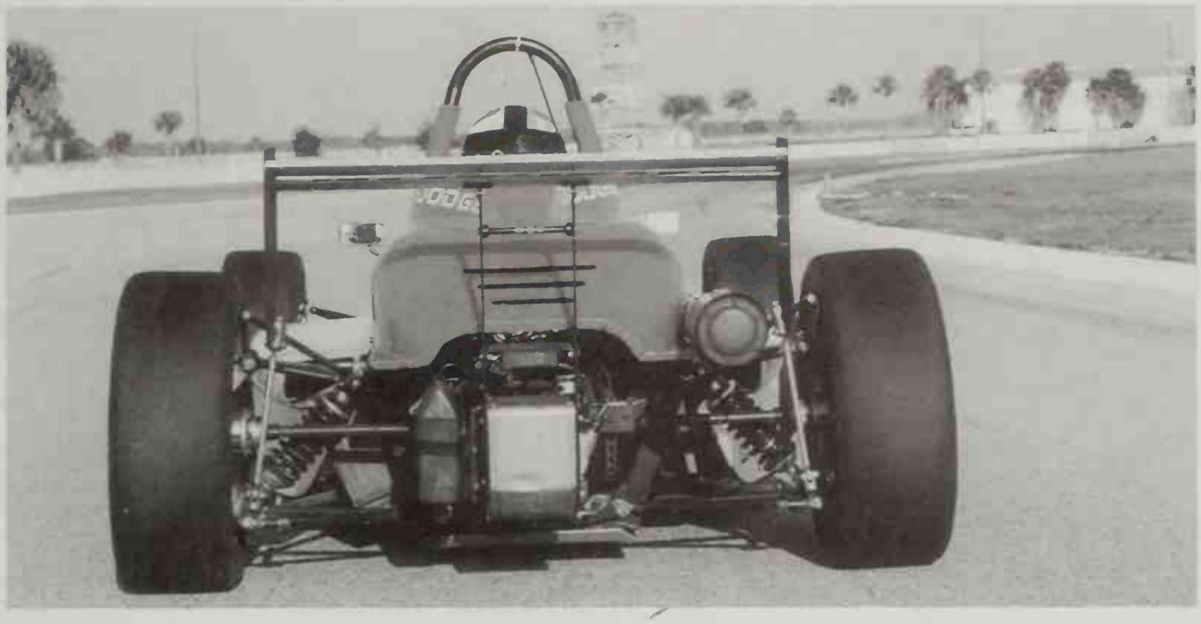

Fig. 7-16. The best path through Turns 4 and 5 will be the one with the least steering input.

Throttle Application. Turn 3 leads onto a long period of acceleration, so once you get the car turned in, you want to get to full power as soon as you can manage it, well before the apex. Wheelspin may be a problem since it's a slow corner taken in a lower gear. Turns 4 and 5 look to be, essentially, an acceleration zone, so you'll expect to have the throttle to the floor except for the upshifts.

Braking. Turn 3 is one of the slower corners on the circuit and, since Turn 2 is fast, there will be a lot of speed loss required here—definitely use threshold braking. Since the corner leads to a long acceleration

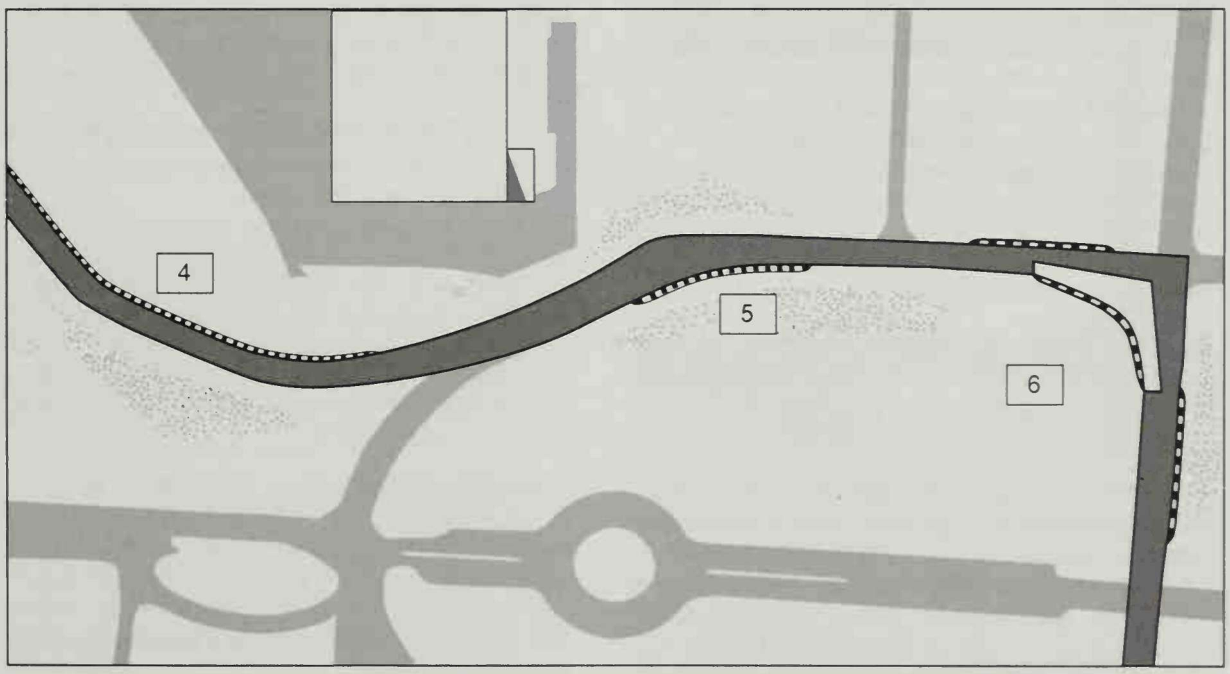

Fig. 7-17. To be on target for Turn 6, there's no need to hit the apex curbing for Turns 4 and 5. Be careful not to drift too far right or, through Turn 4, you'll get into the marbles.

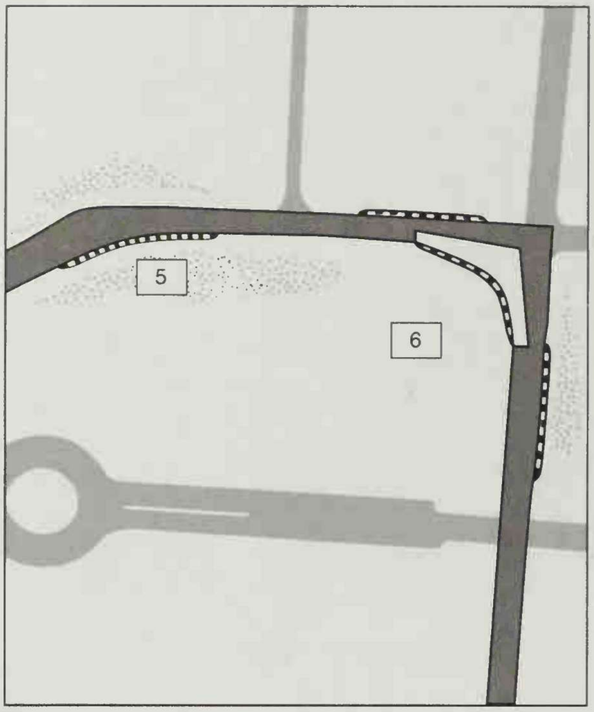

Fig. 7-18. Turn 6 is the first uncomplicated corner on the circuit. The turn-in for 6 may be later than it appears because of the decreasing radius.

zone and it's an increasing radius corner, throttle application will come early, so don't expect to trail the brakes deeply into the corner. Trail-braking about a third of the way or less between the turn-in and apex will probably do.

Experience will tell whether the best approach is to brake diagonally toward the turn-in or to get the car parallel alongside the left edge of the road, depending on the room available, before going to threshold braking. Having the car straight under braking will be the priority, regardless of the direction it's pointing.

Turn 6

The Line. Finally, a straightforward corner. The corner shape has a decreasing radius, so the turn-in may be later than it appears (see Fig. 7-18). Still, it's pretty simple: start on the left, apex on the right, track all the way out on the exit.

You'll need to decide whether you have to drive the car parallel to the edge of the road at the entry or whether a diagonal approach to the turn-in point allows better braking without making the initial turn-in any tighter. You can probably only resolve this at speed. If you let the car go straight past the apex of 5, you can drive straight toward the turn-in for 6 or you can bend the car left, then right again, to align it with the edge of the road for the braking zone for 6. The advantage of doing this would be to make the total amount of direction change in 6 smaller. If you arrive at the turn-in point with the car pointing more toward the left, it just increases the number of degrees of direction change you need to accomplish.

On the other hand, in order to do this left-right swerve to align the car with the edge of the road, you'll be adding tire scrub and perhaps slowing the acceleration up to the corner. Affecting your decision at this point is just how much additional direction change is involved. Is it 45 degrees more, or less than 10 degrees additional turning? In this case the increase is small, under ten degrees. Another factor is whether you can comfortably align the car with the edge and get it straight before the braking point. For now you'll decide to try a diagonal approach to 6 and, once up to speed, try the alternative approach and see which you like better.

There is a very visible two-layered curbing, striped in blue and white, at the corner entry. You can gauge your turn-in on the clearly visible junction between the low and high curb (see Fig. 7-19).

The inside edge of the road bends very gently toward the apex, making the radius of the corner seem extraordinarily large on the way in. You'll have to make a point of ignoring the temptation to get in early.

At the corner exit, two things happen that may have an impact on your line and how the car will feel at speed. As you cross the middle of the following straight heading for the track-out point, you come over the crown of the road onto a section of negative banking (off-camber) and at the same time the track surface changes from concrete to old asphalt. The apex may be later than you think because the car is going to lose cornering grip when it gets to this part of the road.

There's not much of a straight following Turn 6, so you probably won't play with a later apex, hoping to eke out a little extra exit speed.

Should you get in too early, there's a gently angled curb at the track-out that gives you an extra two feet of road and could catch the car in a pinch. It flattens out past the trackout and you could use this flat painted portion but you'd want to hustle the car back onto the racetrack proper. You want to avoid giving the driveline a shock by dropping the left rear onto the sand where the curbing ends.

Throttle Application . Although Turn 6 has the same angular change as Turn 3, it is a much longer corner. The transition to aggressive throttle will have to come later than in 3, or you'll gain too much speed to be able to keep it on the road at the corner exit. Wheelspin doesn't look to be a problem since 6 should be faster than 3.

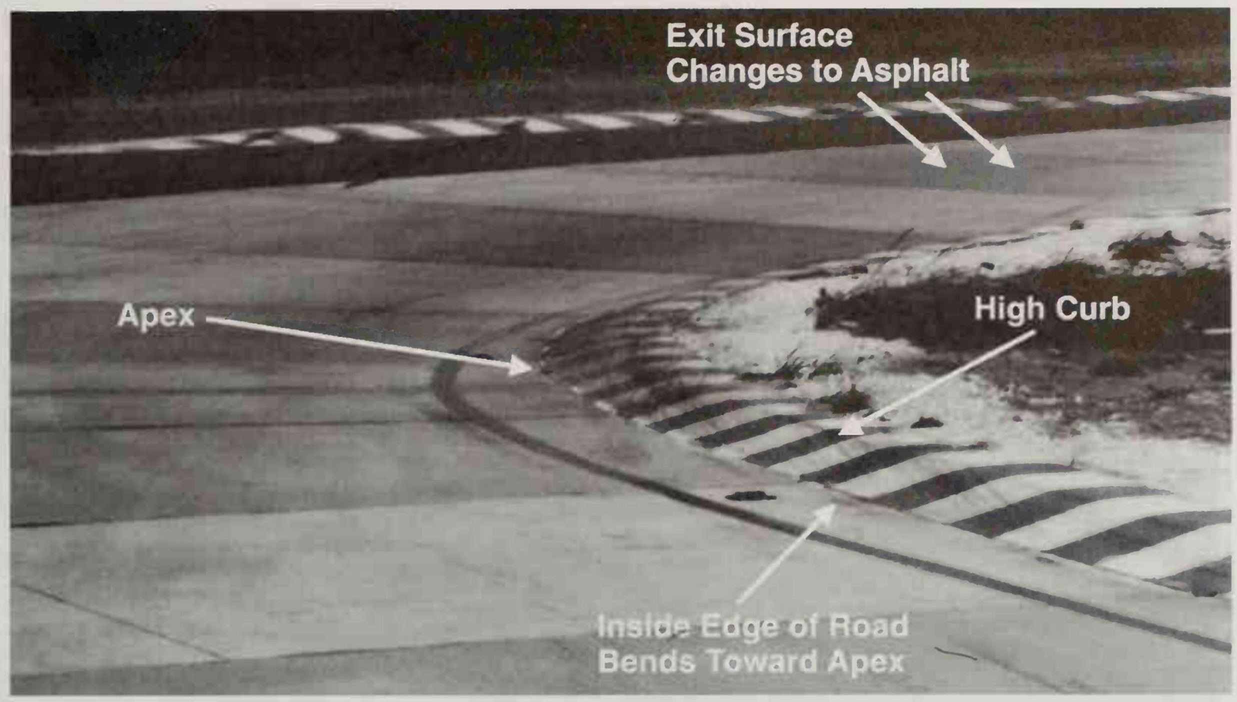

Fig. 7-19. The high curb in Turn 6 makes the apex easy to spot. Watch for surface changes at the exit.

Braking. Coming at the end of a 1/4 mile acceleration zone, the entry speed will be high and, although 6 is not as slow a corner as 3, you can expect to need a lot of speed loss. Threshold braking, or close to it, will likely be required.

Turn 6 is a longer corner than 3 and a decreasing radius corner. It's likely that the throttle application will come on later than in Turn 3 and the braking will go on longer. Expect that you'll trail the brakes deeper into the corner, especially if it helps you carry some of the straightaway speed with you into Turn 6.

Turn 7

The Line. This turn is a textbook right-hander with a well-marked apex curbing (see Fig. 7-20). There is a dark road surface discoloration off the left edge of the road which will help you establish a turn-in point.

There looks to be a good deal of positive road camber (banking) around the apex, so you might be able to add a bit of steering lock to briefly get the car's direction changed more around the apex.

The bad news is that the road goes sharply off-camber as you approach the track-out and, judging from the wide patch of raw dirt to the left of the track-out curb, it forces a lot of cars to go wide on the exit.

It looks as though the curb can be used on the track-out, but there's a deep drop-off on the left that will bottom out the suspension if you drop your wheels in there (see Fig. 7-21).

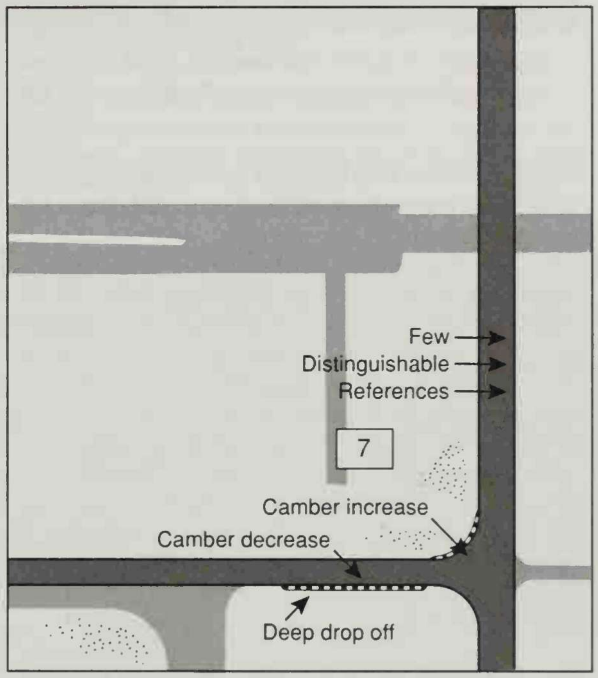

Fig. 7-20. The major difficulty in Turn 7, a textbook right-hander, is its lack of distinguishing features at the corner entry.

Throttle Application. This turn is the tightest 90-degree corner on the circuit, so aggressive throttle should come earlier than the other two. Since it

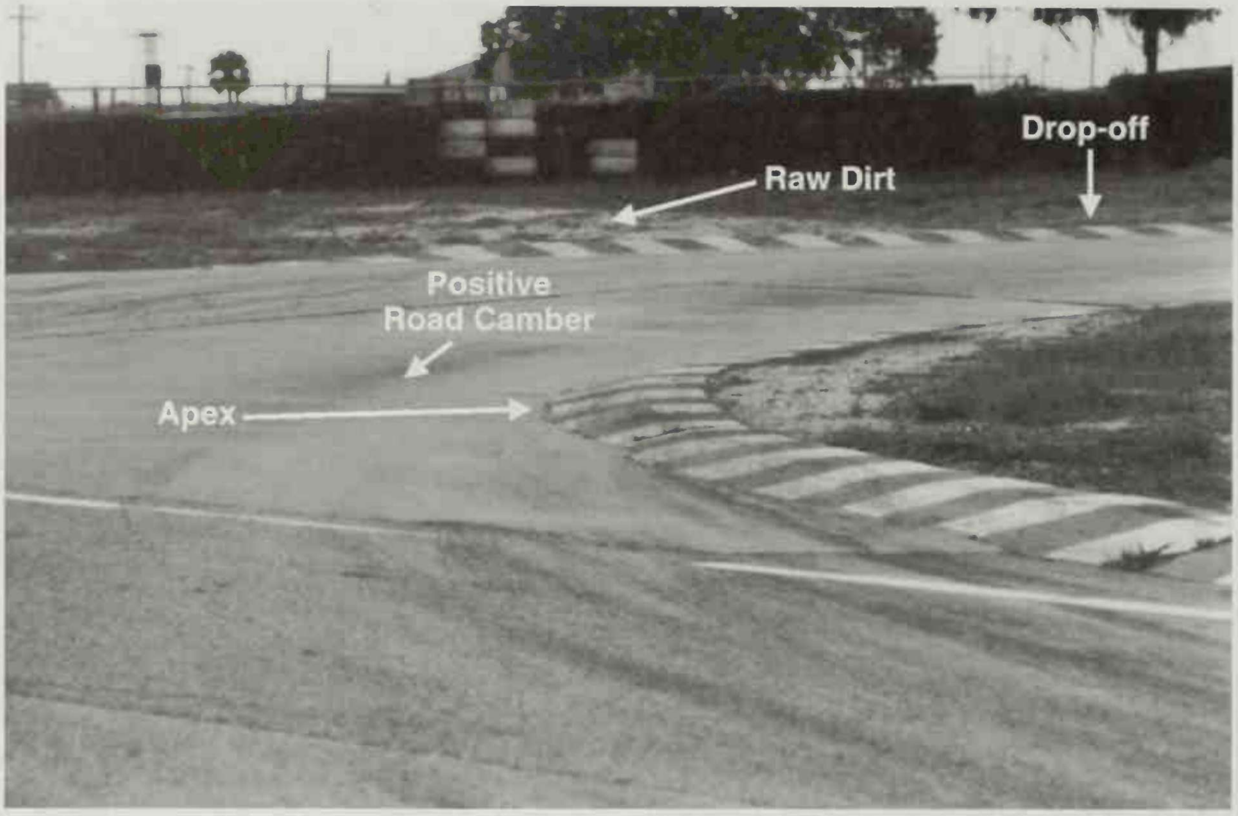

Fig. 7-21. Watch for the deep drop-off at the exit for Turn 7; you can see here that plenty of drivers have gone wide on this turn and landed in the dirt outside the track-out curb.

might, however, be a first gear corner, you might need to be more progressive on the way to full power to avoid wheelspin.

Braking. The approach speed won't be as fast as 6 but the corner sure looks slow and short. You'll probably use threshold braking but not for as long as in 6. The trail of the brakes into the corner should also be considerably less simply because there isn't much distance between the turn-in and apex—even if you trail brake halfway to the apex, it's still less than three car lengths. The biggest problem in 7 is finding a good braking point since there's no turn-in curb at 7 and there are few road features to help you.

Turns 8 and 9

The Line. Here are another pair of corners that are connected, so you'll have to compromise one for the other (see Fig. 7-22). Since Turn 9 leads onto the longest period of full throttle acceleration on the racetrack, there is no doubt that you compromise the path through 8 in order to get to the right side of the road for the turn-in point for 9.

Turn 8 comes just 100 yards after the very slow Turn 7 so it's conceivable that you may not have to slow for 8, but just drive through it up to Turn 9. Only time on the track will tell.

The turn-in will be easy to locate since it comes slightly past where the intersecting road of the 12 Hour course adjoins the Test Circuit. Plenty of reference points here. In order to get you in position for the turn-in for 9, the apex in 8 will be very late, actually at the very end of the striped curbing on the right. It's easy to spot.

The turn-in for 9 is also an easy landmark. There is another curbing on the right near the turn-in, but you can't let it fool you. The actual turn-in point is beyond it, marked by a dark discoloration of the track surface.

The apex, however, is another matter. The corner is long—it's almost 500 feet from the turn-in to the track out—and the car has to change direction about 110 degrees. This means that the apex is almost a football field away when you turn the wheel at the turn-in. The entire inside edge of the corner is lined with a 4-inch high curb so there's no chance to see the apex until you're halfway to it (see Fig. 7-23). Complicating matters is the fact that, although it looks like a long constant radius corner at first, it's actually a long decreasing radius corner and, therefore, requires a later turn-in. It will be very easy to early apex this turn.

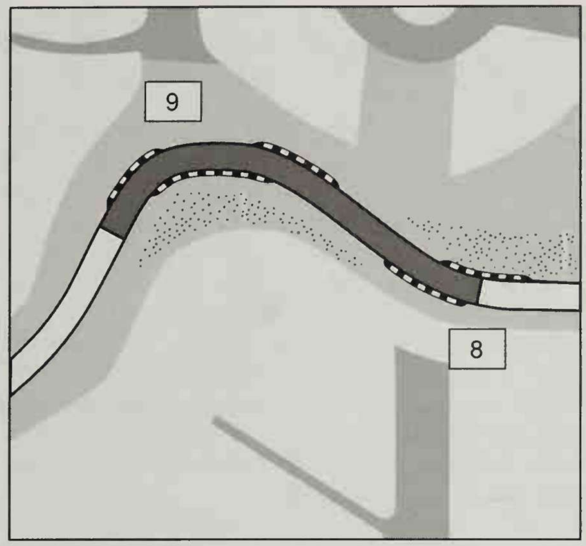

Fig. 7-22. Turns 8 and 9 have plenty of reference points, from the intersecting road of the 12-Hour course at the turn-in for 8 to the turn-in curb for 9. Turn 9 leads to the longest straight on the circuit.

This is the most important corner on the circuit as far as exit speed is concerned. If you get into the apex early, you'll have to lift out of the throttle to keep from running off at the exit; and the last place you want to be decreasing throttle is coming onto the longest straight. It will be crucial to get a reference point so that you can check on your progress toward the apex in time to make an early correction. This is going to take a lot of repetition and a sharp eye for reference points that coincide with the correct path.

Throttle Application. You won't be able to tell until you get out there, but it looks like you might be able to drive through 8 without lifting. If not, you want to plan to lift just before turn-in and squeeze the power back on as you turn in to stabilize the car. You don't want to turn in with your foot on the floor, then lift while the car has heavy cornering loads on it because TTO will get the tail out.

Other than the sweepers, Turn 9 is the longest corner on the racetrack, with almost 500 feet between the turn-in and track-out. Even though you would like to go to aggressive throttle as early as possible in order to maximize the exit speed onto the longest straight, you'll have to be patient here. If you go to a lot of power early—say, in the first 50 feet following the turn-in—the car will gain too much speed by the apex and you'll almost certainly have to lift coming off the corner to keep the car on the track. The throttle application is likely to be a long progressive squeeze toward the floor, being constantly sensitive to whether your throttle application is forcing you off-line. You still want to be hard in it before the apex, but not 100 feet before the apex. Even low-powered cars are likely to take some delicacy here.

Braking. If you do any braking in Turn 8 at all, it should be just a touch before the turn-in.In Turn 9 it looks as though you will be able to get the car straight

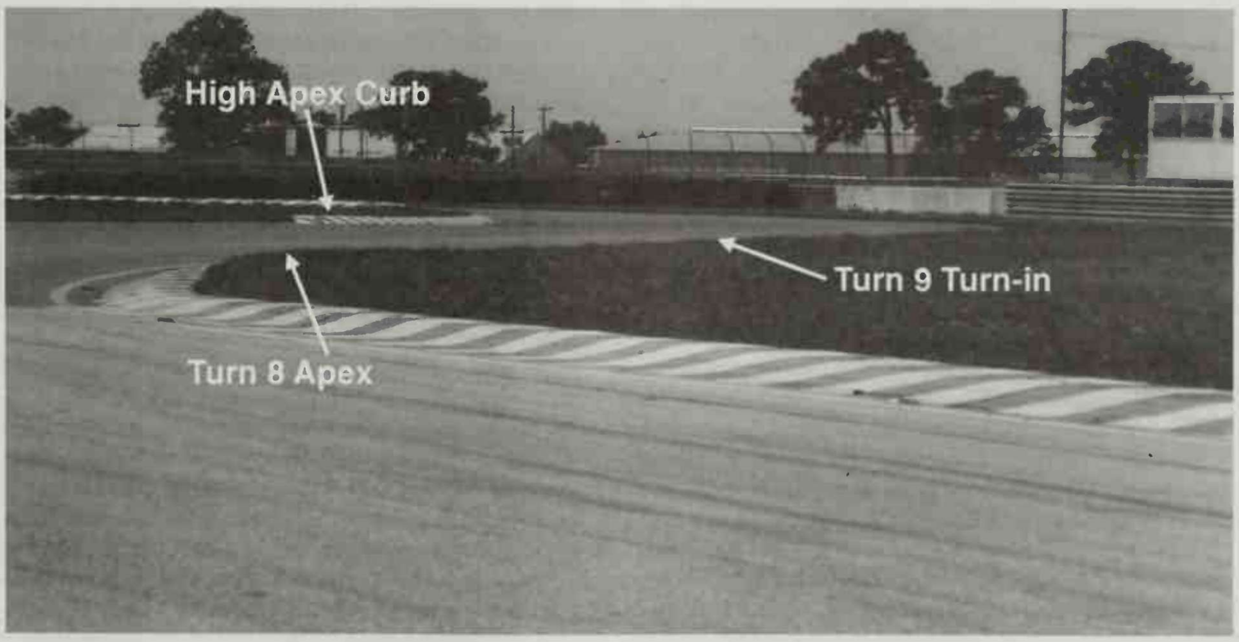

Fig. 7-23. Exit speed is crucial in Turn 9, so you'll compromise 8 for 9, apexing late through Turn 8. Watch for the apex curb in Turn 9, which is hard to spot until you're halfway there because the curb is so high.

long enough at the exit of 8 to begin braking in a straight line for 9. If not, you'll have to focus on taking a line through 8 that makes it possible. It appears that only sloppy car placement in 8 will cause a problem here.

Considering the fact that Turn 9 is such a long corner and that aggressive throttle will come on a long way past the turn-in point, you might be able to make up time at the corner entry by decelerating and turning toward the throttle application point. This is especially true since Turn 9 is a decreasing radius corner. You may want to use the big radius in the early part of the turn to carry extra speed in, as long as it doesn't interfere with the throttle application. For this corner, there might be a lot more brake-turning than straight-line braking.

Turn 10 : The Hairpin

The Line. Turn 10, or "The Hairpin" (see Fig. 7-24), is a classic corner, one driven by motor racing's greats, from Moss to the Andrettis. It is not, however, terribly complicated. It will be a standard out-in-out corner, but since it is both slow and leads onto a substantial straight, you may have to turn later and apex later than it appears. This is especially true since there's a big direction change involved—you change the car's course over 135 degrees. There aren't many corners around that are this slow, yet seem to go on this long.

Reference points are plentiful in this turn. The concrete starts 50 feet short of the turn-in and each

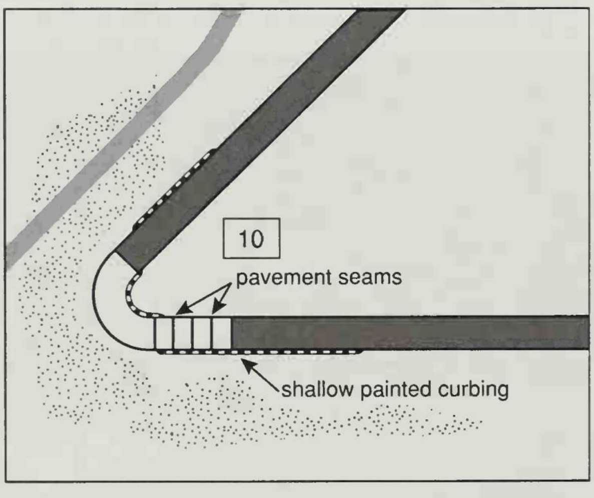

Fig. 7-24. Reference points, especially for braking, are crucial in Turn 10, The Hairpin, since it is the slowest corner on the circuit. The change from asphalt to concrete will be especially helpful for judging your braking point and turn-in.

12-foot section has a seam which can be used for a reference. In addition, there are small square reflectors, useful at night in the 12 Hour, cemented to the edges of the racetrack.

The apex curb, striped in red and white, has a red dot painted on the track surface which can be an aiming point for the apex. The problem is that the dot is so far around the corner that you can't see it from the turn-in. The initial turn of the steering wheel is toward a point beyond your vision so it will help to develop a waypoint—some reference on the track surface—to let you know whether you're inside or outside the proper path (see Fig. 7-25).

There is no road camber to speak of and, although the concrete paving stops 80 feet or so short of the track-out point, it shouldn't be a major traction loss problem like Turns 6 and 7 are.

The concrete track-out curbing is absolutely flat and may as well be part of the racetrack. Of course you'll use it, but it doesn't extend as far down the track as you'd want it to, so there will have to be a little jog to the right to rejoin the racetrack proper, just past the track-out.

Throttle Application. This turn is the slowest corner and the one with the biggest angular change. Since it's followed by a straight almost a quarter mile long, exit speed is important. The corner is very slow, most likely first gear, so wheelspin can be a very real problem. You'll have to squeeze the throttle on, trying to keep a reasonable yaw angle going by feeding on just the right amount of power to keep the car sliding in a range where the tires' slip angles are best. Since the corner is so slow and you have the wheelspin capability of first gear, it's a corner where you can have a lot of fun with big, lurid tail-out slides. Remember to restrain yourself. The key is getting the power down in order to get a good shot onto the straight.

It's going to be easy to get obsessed with getting the power on early here. If you get on the power too soon, before the car gets to rotate into a usable yaw angle, you're likely to trip it into understeer, which you'll have to solve with a breathe or lift late in the corner.

Braking. The approach to The Hairpin can be a little intimidating. A huge billboard is right in your line of sight as you approach the corner and at speed, it seems like it's one foot off the racetrack. A brake failure here would not be good, as your options are limited. There is a makeshift escape road to the left which you could take to avoid driving straight into the tire barrier, but it wouldn't buy you that much time. You should make a mental note to be sensitive to the onset of brake fade in any of the other corners and plan that if you sense it, even a hint of it, you'll give the brake

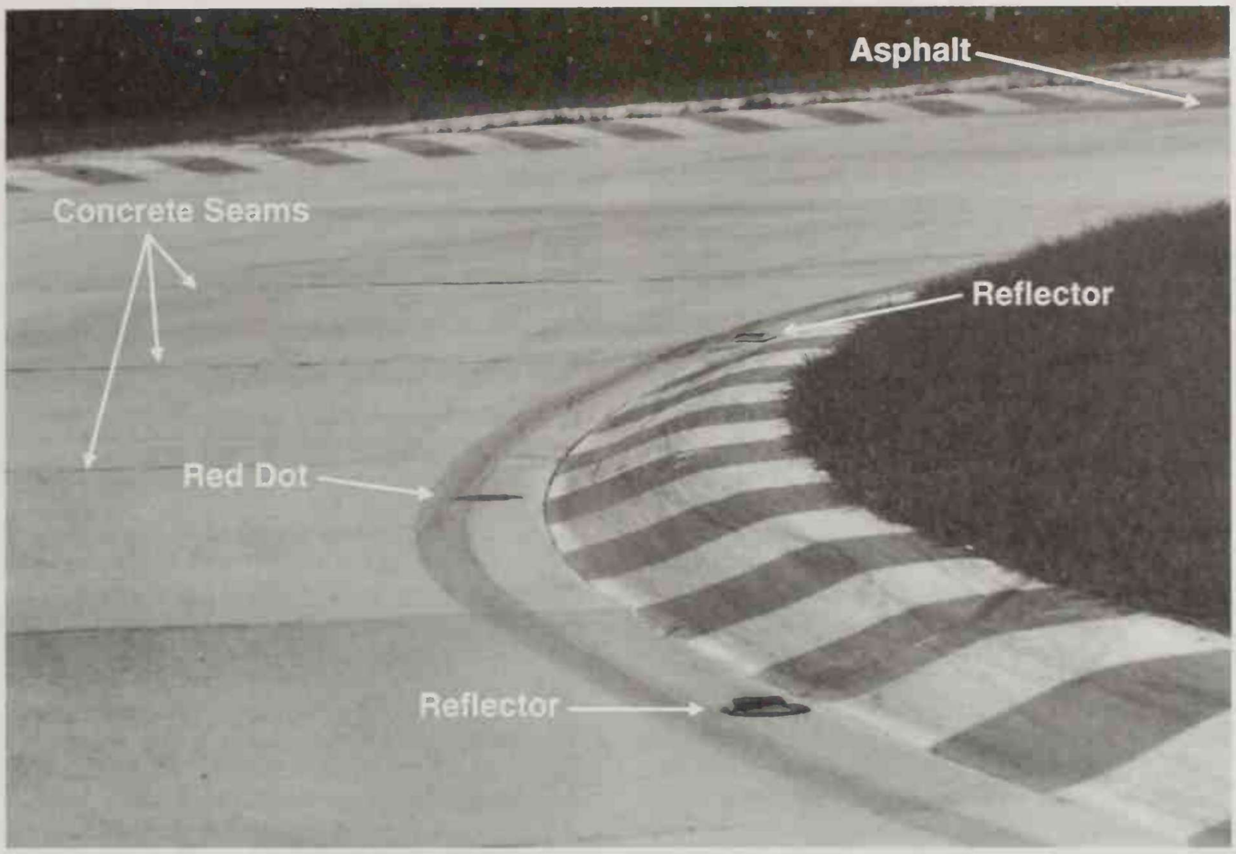

Fig. 7-25. The seams in the concrete section of The Hairpin serve as good reference points. There is also a red dot painted below a white section of the apex curb which, along with reflectors on the edge of the track, help to establish your apex once you've made the turn-in.

pedal a tap or two before the brake point for The Hairpin, just to make sure somebody's home.

If there ever was a threshold braking corner approach, The Hairpin is it. You'll have to go from however fast the car will go—around 110 m.p.h. in our Formula Dodge race cars—to as slow a corner as you'll ever see. Braking points are simple to pick out as there are either number boards or pylons set up at 100 foot increments from the turn-in. Expect to use threshold braking and to work the braking point toward the corner as you practice. Remember that getting to the turn-in point at the right speed is just as important as doing a perfect threshold braking level. Especially in the early sessions on the racetrack, don't be shy about modulating out of maximum braking in the last portion of the braking zone to increase your speed at the turn-in, getting a feel for what turn-in speeds are possible.

Remember that in a tight and long corner such as this one, it's easy to pick up gobs of understeer by going to the throttle before the car has rotated to a usable yaw angle. Continuing the braking past the turn-in, then using a quick release of the pedal pressure, can help get the tail out to where you want it: then it's a matter of holding it there with throttle. Remember, too, that a pause between the relaxation of brake pressure and the application of throttle can also help generate yaw angle.

Gearing

As you may have noticed, we've avoided recommending gear selection for each corner. We've done this because you really won't know which gear is appropriate until you try it. In practical terms, you pick a gear you think will get you from the throttle application point to the track-out without having to shift to another gear. As you first try the course, you'll take your best guess and probably be off by, at the most, one gear. You'll either have guessed a gear too low, in which case the engine will reach the redline before the track-out, or too high, in which case the engine labors through the corner in a rpm range below the best power range of the engine. You adapt by trying a better gear next time.

Most of the corner entries are relatively small speed losses, so you can expect that the downshift will be to the next lowest gear and that it will be a standard double clutch downshift. If possible, you want to do the downshift while the car is going straight on the approach to the corner in order to minimize the chance that a mismatched downshift will lead to a spin. In the corners where you may try to carry extra entry speed—Turns 6 and 9—it is possible that the car might be going too fast at the turn-in point to do the downshift before turning in. In these corners, you'll have to make a note to try to do the smoothest downshift you can manage to keep from upsetting the car.

The Hairpin, however, opens up the possible choice of skipping gears and perhaps choosing to do a fourth-to-first downshift. If you decide to choose this option, you have to remember to delay the shift to a point where, when the clutch comes out in first, the car is going slow enough so that you don't overrev the engine or break the rear tires loose by engine braking.

All Set To Go

If we all went through the preceding analysis of the racetrack before setting out to do battle with the stopwatch, we'd have many fewer surprises in store. You would know, by walking your way through it, which corners are the most important and are therefore most deserving of your attention. You would know which corners are straightforward. You would know which corners are going to require a majority of threshold braking and which will require a more subtle touch. You would begin to have an idea of which corners are going to call for quick, point-and-squirt type throttle application and which corners may require long, well-modulated control of the accelerator.

Before you set out, you want to have a plan for each corner. Suiting up, jumping into the car and going out on the course, hoping your "natural ability" will solve all your problems, is just dreaming. Knowing what you want to do comes first.

TIME TO DRIVE

Having gone through the off-track homework, it's now time to apply it in the race car and begin the task of become familiar with the course in the car you're going to pace. I'll take you with me through a lap at Sebring: once you're in your equipment and buckled in the car, it's time to go.

I take a good look in the left hand mirror to see if anyone is coming down the pit road, slip in the clutch, and we're underway. The pit lane at Sebring is wide and uncluttered, unlike many pit lanes, but I still take it easy going out, since there are a dozen or so drivers lapping Formula Dodges here today and there is a small crowd milling around the cars near the pit exit. Once clear of the crowd, I accelerate up to 5,000 rpm in first, about 50 m.p.h., before slowing and looking to the left at the pit exit for traffic—a simple street driving action that a surprising number of new racedrivers seem to forget.

The water temperature is coming up on 160 degrees so I can feel free to use the full 6,000 rpm redline without fear of stressing the motor. I enter the racetrack at Turn 4 after having shifted from first to second

"It's all with a person just having a little session with himself going in. Most times, the less experience you have, the more unknowns are going to come at you when you're in there. There's going to be so much confusion and that's when a mistake happens. You just—gosh— its happened.

"So that's the time when you figure, 'I'm going to go in there and I'm going to build up to it. I'm going to go a little bit—the heck with what they're saying back there with the stopwatches—I'm just going to go out there and do it at my own pace.'

"If nothing else, it shows a lot of maturity. You'd be surprised how much of that is respected back in an area where you think you're being criticized for being slow. To be patient, slightly patient with yourself, I think, is something that will always work for you."

—Mario Andretti

Fig. 7-26. Turn 6 brake point: I start at the middle of the racetrack between 5 and 6. It will be later as I progress. Don't over-slow. Brake reasonably hard but not like in 3. Use constant-level brake turn so as not to over-rotate. Turn-in: 15 feet past the change in curbing height, the fourth blue stripe on the curb, as a start. The curb is relatively flat and it doesn't look like touching it will upset the car.

Fig. 7-27. Turn 6 apex: yellow line on inside of track turns black at apex and this coincides with a left to right track seam with a change in pavement color. Track out: the very end of the steeper curbing.

Fig. 7-28. Turn 7 brake point: the cone is helpful but the disrupted turf is as easy to see. Trail out of brakes quickly but avoid popping off. Turn-in: the last reflector and some darker eroded pavement on the left. Keep the left tires to the left of the heavy marks.

at 6,000 rpm (60 m.p.h.) and accelerate to 6,000 in second (75 m.p.h.), entering the racetrack proper at the apex of the right-hand sweep of Turn 5.

I'm accelerating smoothly, squeezing the throttle toward the floor in a much more gradual way than I will eventually, but this is a warm-up session, so I can afford the luxury of floating gently around the course. I do an easy upshift to third and accelerate at half throttle up to the entry of Turn 6. I approach the left

Fig. 7-29. Turn 7 apex: I notice that a blob of paint from the apex stripe extends onto the track surface and the stripe itself is darker than the rest. Track out: again, the steep curb turns flat. I'll be able to use the flat section later.

Fig. 7-30. Turn 8 brake point: don't need one. Turn-in: 5 feet past where 12 Hour curbing ends. Look right.

side of the road on a diagonal from the apex of 5 and the change in the two levels of curbing, which I thought would be a good turn-in reference, comes clearly into view. Even on this prelude to a warm-up lap, I'll try to run the racing line and look for reference points to guide me around.

The braking for Turn 6 is light, since the brakes are cold and the first application begins the process of warming them up. I do a quick downshift to second and find that the motor blips eagerly, which is not always the case.

As I approach the curb, I divert my eyes toward the apex, picking up the turn-in using peripheral vision. The steering comes in smoothly; I relax the brakes and pick up the throttle, gently. The arc is right out of 6 and the car naturally uses all the road coming out.

Rather than shift up to third on the short straight between 6 and 7 (the right gear for this section at-speed), I stretch second up to Turn 7, the tight 90-degree right. I get on the brakes early at 7, continuing the warm-up process. The brakes go on easy and early and I wait two seconds as the car decelerates toward the corner before doing a double clutch downshift to first gear.

As I approach the turn-in point, I divert my eyes toward the apex so that when I turn the wheel, I'm looking in the direction I am about to go. At the turn-in point (this track stain is also clearly visible), I turn the wheel to the right, ever so slowly on this warm-up, and the car bends into the turn.

Now I breathe back the braking level and when it's at zero, start squeezing on the throttle. It all feels like slow motion now. The car scoots away from the corner and I shift up from first to second ten yards past the track-out point.

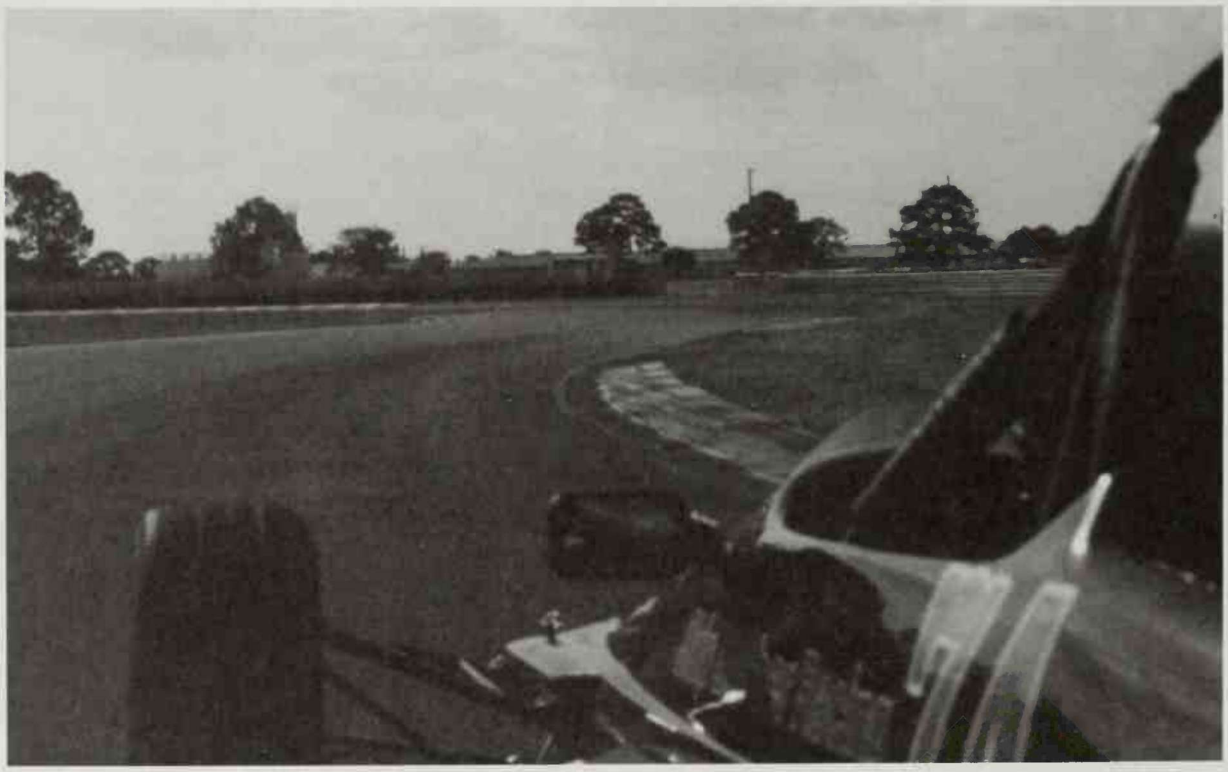

Fig. 7-31. Apex of 8: aim a little past the end of the curb. Brake point for Turn 9: start halfway between the two curbs. Try smooth, light brake turn. Delicate downshift while turning. Pick up part throttle after trail. Turn-in for 9: discoloration (looks like sealer) 10 feet past the curbing.

I decide to get some temperature in the brakes before we get there. As the car is accelerating up to Turn 8, I move my left foot under the steering shaft and put it on the brake pedal. Initially, I don't push on it very hard—I don't want to shock-heat the braking system, I just want to coax it up to operating temperature. It only takes five seconds or so to go from 7 to 8 and I drag the brakes up to the corner, turning right toward the entry of Turn 9, the carousel leading onto the main straight.

Approaching 9, bending to the right on the way to the turn-in point, I lock up the right front tire. Damn. I didn't want to do that! I'm trying to treat the car well by warming it up and in the first minute of the day I lock up a front, potentially flat spotting it. Fortunately, the street radials fitted to the car are reasonably hard and the total weight of the front corner is only around 160 lbs, so the flat spot, if any, will be small.

As I accelerate out of Turn 9, I keep it in second at about three-quarter throttle, still squeezing on the brakes (more lightly now), dragging them all the way down the .4 mile acceleration zone leading to Turn 10, the Sebring Hairpin. Next lap, we'll be accelerating through the gears on this long sweep to the right. There will be very little time when the car will be going dead straight. As I drift toward the left verge, passing under the bridge, I'll get the car straight for just 2 seconds or so before applying the brakes for the slowest corner on the circuit.

I'm just cruising up to The Hairpin now at about 50 m.p.h. I move my left foot back over to the clutch side of the footwell, brake the car conventionally with my right foot, and do the downshift to first to get a running start out of The Hairpin for the start of the first data collection lap.

Fig. 7-32. Apex of 9 impossible to see. Look as far ahead as possible to inside of road. Hold car right of track seam until beginning of exit curb is visible. Correct, if necessary, in order to guide car to inside.

Fig. 7-34. Straight approaching Turn 10: Hold the car along the right edge until the pavement change, then track left (after checking mirror).

Fig. 7-33. Turn 9, first 1/3 of flat exit curb. Aim just past the change in steepness of the curbing.

Fig. 7-35. Turn 10 brake point: start with second pylon. Save threshold for later.

Fig. 7-36. Turn 10 turn-in: last reflector on concrete. Very black, just 12 feet before the white line turns right.

Fig. 7-37. Turn 10 apex: aim far around corner. Think of trying to come around behind the curb you can see from the turn-in.

I accelerate away from the apex of The Hairpin and get full throttle before the exit. The motor feels good and peppy. The tach needle moves quickly in first gear and I let it go to 6,000 before shifting to second. Foot to the floor in second, I'm accelerating through about 70 m.p.h.; then I shift up to third at 6000 and put my foot back to the floor.

I'm approaching Turns 1 and 2 at 82 m.p.h., and there's nothing fancy about my braking point. I use my judgement (biased toward the conservative) and squeeze on the brakes lightly, slowing the car to 72 m.p.h. by the turn-in point. Turn 1 is a gentle bend, so the steering wheel motion is small and, at this speed range, there is no perception of sliding as the car solidly tracks toward the apex. I carry the light braking past the turn-in point and pick up 50% throttle to keep the car comfortable and balanced through the right-hand portion of this right-left sweep.

Fig. 7-38. Turn 10 apex: Once halfway across the straight pick up the second reflector.

Fig. 7-39. Turn 10 track out: use all flat curb, sneak back at very end.

I'm down to 74 m.p.h. as I transition the car from turning right to turning left, trying to do this fluidly; easing the steering wheel away from right lock, then easing in steering effort to the left. I can see it's going to be hard switching my vision away from the turning point for 2 over to the apex. It's going to have to be done quickly, especially as we go faster, to get a good sight on 2's apex.

By the time I'm headed for the apex, the throttle is at 70% and we're picking up speed, reaching 80 m.p.h. in third gear at the track out point. All this is solidly under control, no sliding around, no sudden twists of the steering wheel or jabs at the pedals.

As I proceed around the course I start to pick up my reference points, based on what I discovered on the early drive-arounds in a street car. The Formula

Fig. 7-40. Turn 3 brake point: braking pylons here are very helpful. Start at first one, work it down later. Brake reasonably hard, trail it a quarter of the way in. Turn-in: start 5 feet before the striped curb starts.

Fig. 7-41. Turn 3 apex: seam in concrete goes diagonally left to right. Apex where it meets the inside curb. Track out: use all the road, left tires touching curb.

Dodge, in which my eye level is two feet closer to the ground than the Intrepid sedan I first drove around the course, changes the perspective. I look for the brake points, turn-ins and apexes which I scouted earlier and use most, but in the process discover others.

The mental process is one of looking forward, almost straining to see your next marker as you go around the course. This line development goes on lap after lap, always looking for better references and establishing what works best with this car on this day. The work—the challenge of pursuing perfection—is the fun part.